Project List Fall 2025

Graph Theory

Total Marks: 10 (5 Marks Code + 5 Marks Presentation)

Submission Deadline: **1 December 2025**

Programming Language: **Python**

Recommended Libraries: `networkx`, `numpy`, `pandas`, `matplotlib`, `scikit-learn`, `node2vec`, `pyvis`, `pytorch-geometric (optional)`.

General Instructions

- Each group must choose **one project** from the list below and fill in the online sheet.
- Datasets should be either from the given links or freely chosen.
- Final submission should include:
  - Source code (Python file or Jupyter Notebook)
  - A short presentation
- All submissions are due **before 1 December 2025**.
- Total Marks = **10 (5 for Code + 5 for Presentation)**.

Project Topics

1. **Community Detection in Social Networks**
   Concept: Identify user communities using modularity optimization or Louvain method.
   Dataset: Facebook Social Circles (SNAP)
   Roadmap: Build graph → Apply community detection → Visualize clusters.
2. **Fake News Propagation Analysis**
   Concept: Study how fake vs. real news spread differently using graph modeling.
   Dataset: FakeNewsNet
   Roadmap: Create user-post graph → Simulate propagation → Compare metrics.
3. **Graph-based Movie Recommendation**
   Concept: Use a bipartite user–movie graph for link prediction-based recommendations.

Dataset: MovieLens 100K
Roadmap: Create bipartite graph → Compute similarity → Recommend items.

4. **Graph Clustering via Spectral Methods**
   Concept: Apply eigen decomposition on Laplacian to find natural clusters.
   Dataset: Synthetic graph (`networkx.karate_club_graph()`)
   Roadmap: Compute Laplacian → Perform spectral clustering → Visualize clusters.

5. **Traffic Network Prediction**
   Concept: Model roads as weighted graphs; predict traffic intensity using ML.
   Dataset: METR-LA Traffic Dataset
   Roadmap: Create weighted graph → Extract features → Predict congestion.

6. **Graph-based Fraud Detection**
   Concept: Detect anomalous transaction nodes using connectivity and ML-based methods.
   Dataset: Elliptic Bitcoin Transactions
   Roadmap: Build transaction graph → Extract metrics → Apply anomaly detection.

7. **Image Segmentation using Graph Cuts**
   Concept: Apply min-cut/max-flow algorithm for object-background segmentation.
   Dataset: COCO Sample Images
   Roadmap: Represent pixels as graph → Apply graph cut → Display segmented image.

8. **Centrality-based Influencer Ranking**
   Concept: Identify top influencers in a network using centrality measures.
   Dataset: Twitter Ego Network (SNAP)
   Roadmap: Compute centrality → Rank top nodes → Visualize.

9. **Graph-based Feature Selection**
   Concept: Represent features as graph nodes and select central ones for ML tasks.
   Dataset: UCI Iris Dataset
   Roadmap: Correlation → Graph creation → Centrality → ML model.

10. **Graph-based Text Similarity**
    Concept: Represent documents as word co-occurrence graphs and compare similarity.
    Dataset: BBC News Dataset
    Roadmap: Build graph per document → Compute overlap metrics → Rank similarities.

11. **Disease Spread Simulation (SIR Model)**
    Concept: Simulate disease propagation through a contact graph.
    Dataset: Synthetic network generated via `networkx`.
    Roadmap: Create random graph → Implement SIR → Visualize infection over time.

12. **Graph Neural Network for Node Prediction**
Concept: Train a simple GCN for node classification.
Dataset: Cora Graph Dataset
Roadmap: Prepare graph → Implement GCN → Evaluate results.

13. **Knowledge Graph-based Question Answering**
Concept: Build a small knowledge graph and query via similarity search.
Dataset: Freebase Sample
Roadmap: Create knowledge graph → Query via embeddings → Return answers.

14. **Graph-based Email Spam Detection**
Concept: Represent email sender–receiver network; detect spam using graph metrics.
Dataset: Enron Email Dataset
Roadmap: Build communication graph → Extract metrics → Train classifier.

15. **Power Grid Stability Prediction**
Concept: Model power network as a graph; predict node failures.
Dataset: UCI Power Grid Dataset
Roadmap: Create graph → Compute topological features → Predict failures.

16. **Academic Collaboration Network**
Concept: Represent co-authorships and find influential researchers.
Dataset: DBLP Collaboration Network
Roadmap: Create co-author graph → Compute centrality → Identify top authors.

17. **Shortest Path Routing Optimization**
Concept: Use Dijkstra's algorithm and ML to predict faster alternate routes.
Dataset: OpenStreetMap Sample
Roadmap: Build weighted graph → Compute shortest path → Predict optimized routes.

18. **Graph-based Sentiment Analysis**
Concept: Represent words and relations as graph; classify text sentiment.
Dataset: IMDB Movie Reviews
Roadmap: Build word dependency graph → Extract features → Train sentiment model.
